



LCLS-II-HE

LCLS-II-HE vCM Test Results: Newly Developed N-Doping Treatment and Plasma Processing,

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NAPAC2022
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SLAC NATIONAL
ACCELERATOR
LABORATORY

 **Fermilab**

 **Jefferson Lab**



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FRIB

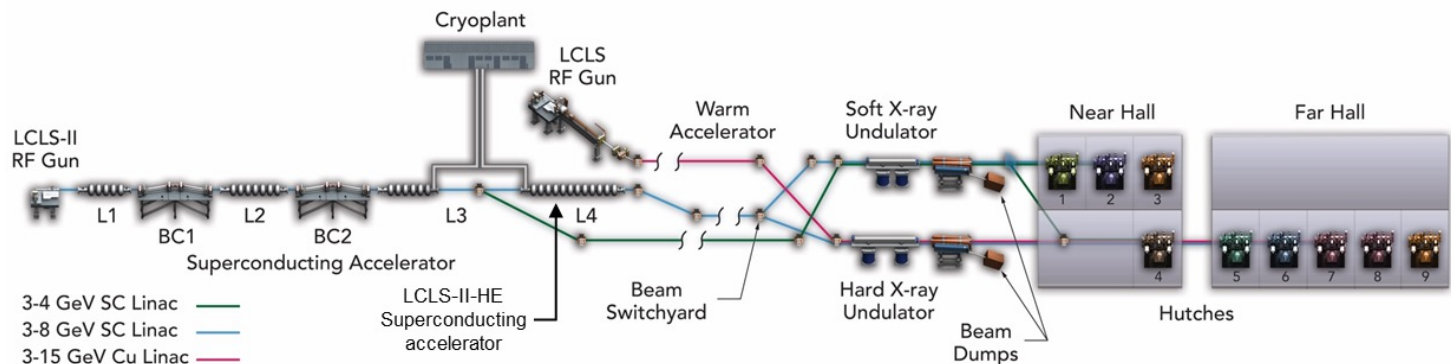
Outline

- LCLS-II-HE recipe development for cavity production
- Verification Cryomodule performance
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LCLS-II-HE

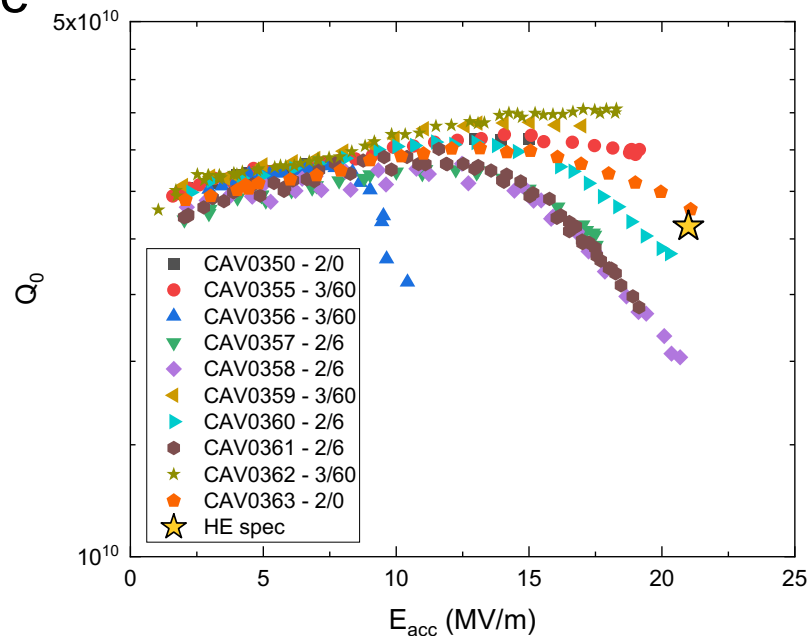


- 24 additional cryomodules to expand the SC linac and increase the energy reach to 8Gev
- The new CMs should operate at 21MV/m with $Q_0 \geq 2.7 \times 10^{10}$
- To meet the requirements:
 - Develop new cavity recipe to maintain high Q_0 and increase quench field $\left. \begin{array}{l} \bullet \\ \bullet \end{array} \right\} \begin{array}{l} 2/0 \\ 3/60 \end{array}$
 - Investigate LCLS-II cavity production processes to reduce performance spread

First 10 LCLS-II-HE R&D cavities arrived at Fermilab

All 10 initial cavities performed poorly

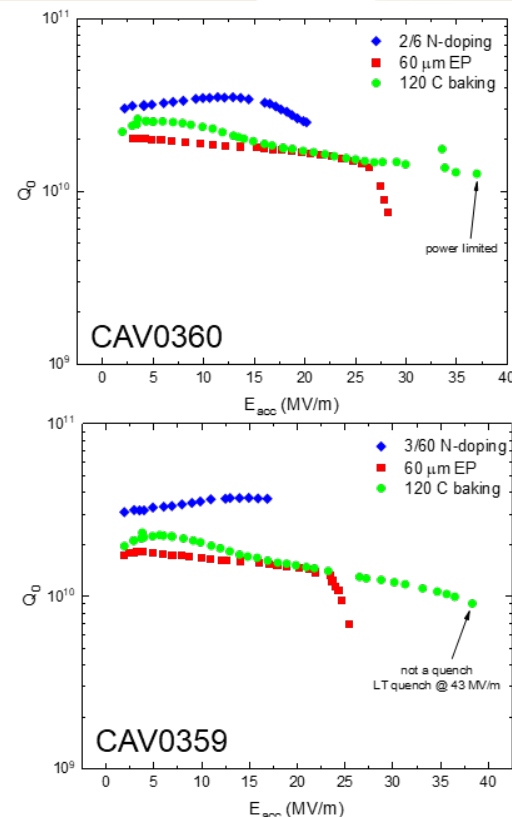
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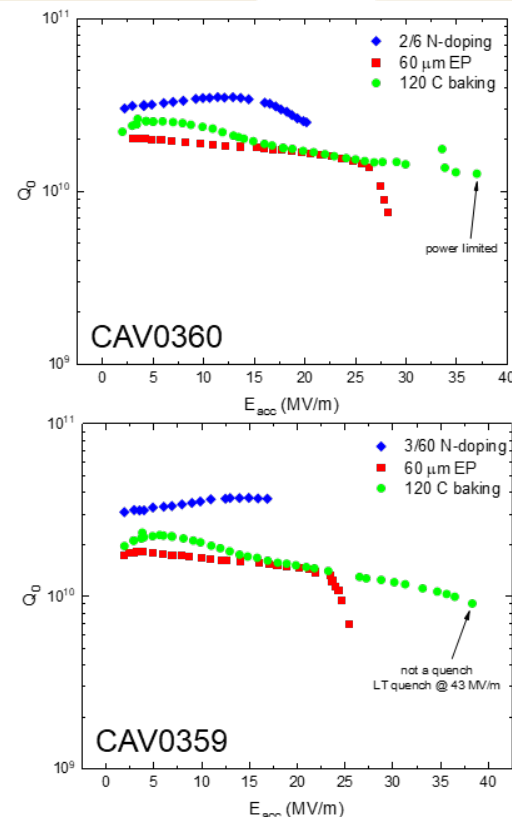
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- 4 cavities were reset with 60 μm EP + 120 C for 48 h and retreated with 2/0 or 3/60
- All the cavities were able to reach fields higher than 37 MV/m!



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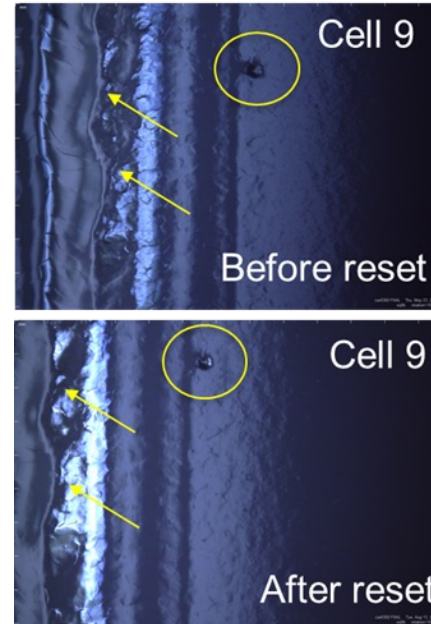
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- 4 cavities were reset with 60 μm EP + 120 C for 48 h and retreated with 2/0 or 3/60
- All the cavities were able to reach fields higher than 37 MV/m!
- What may have caused initial poor performance?



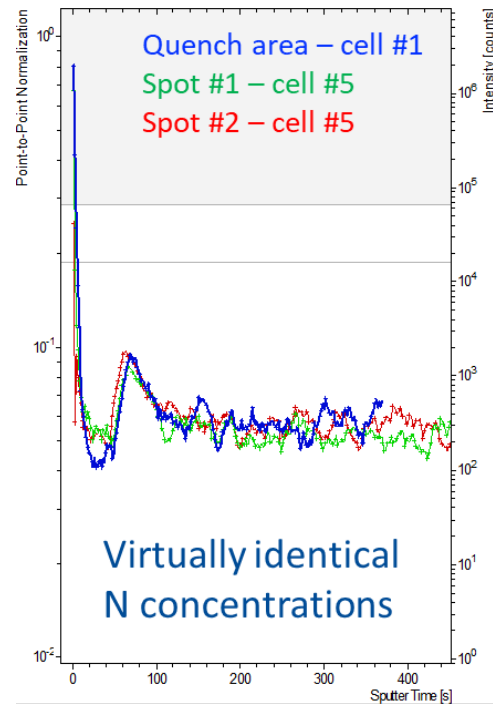
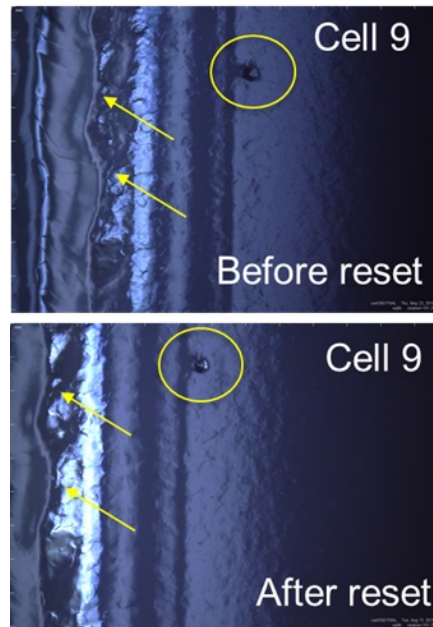
Possible causes for initial cavity performance

- Comparison of optical inspection before and after reset shows: smoother surface but still same surface defects present.



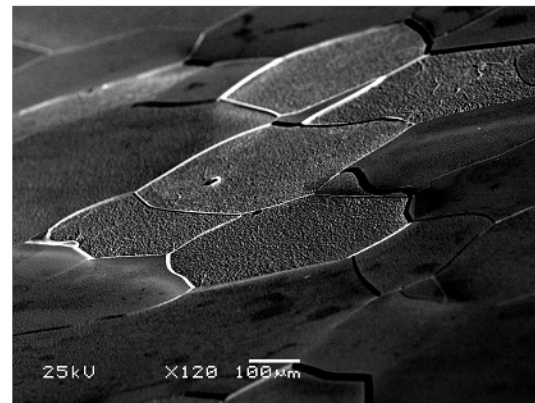
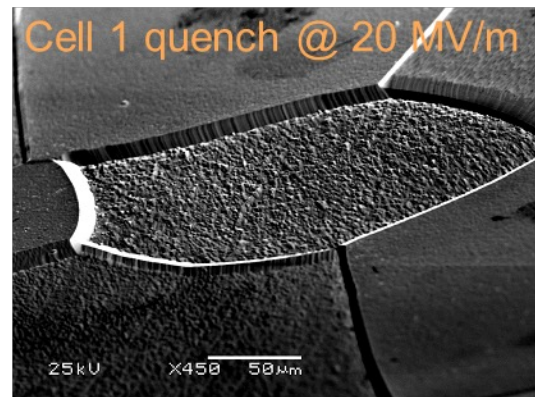
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- Comparison of optical inspection before and after reset shows: smoother surface but still same surface defects present.
- Nonuniform Nitrogen concentration on cavity surface after doping?
 - N concentration is uniform across cavity length after the EP
- SEM analysis shows preferential etching on some grains in the cell of lowest quench value
 - driven by higher number of nitrides in cell1?
 - Final EP of this cavity was done at higher T compared to EP on single cells at Fermilab



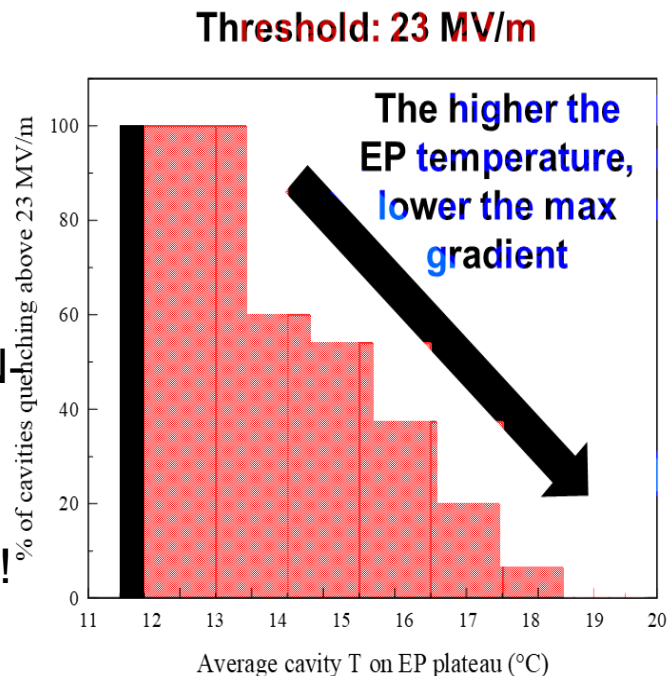
LCLS-II production data: quench field vs EP temperature

Analysis performed on EZ final EP data, LCLS-II production data (reworked cavities excluded) and initial LCLS-II-HE prototype cavities was considered

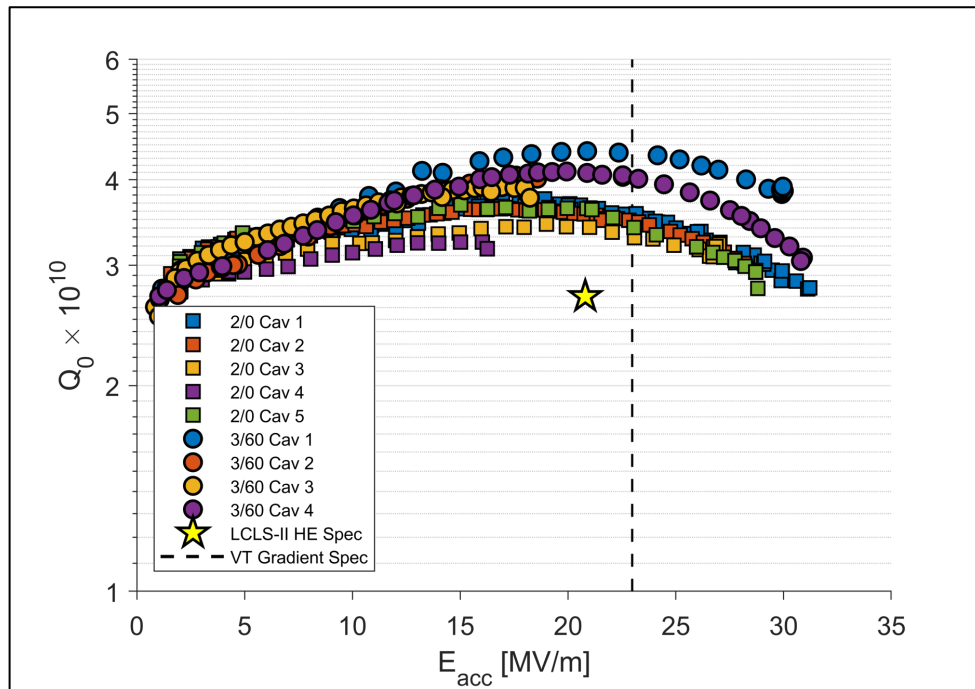
- LCLS-II cavities were EP'd at lower T and quench higher

How does the final EP affect the quench field of N doped cavities?

- At LCLS-II operational field no specific trend
- BUT at HE operational field the trend is clear!
- **To achieve higher field low EP T is necessary**



New recipes + cold EP at the partners Labs

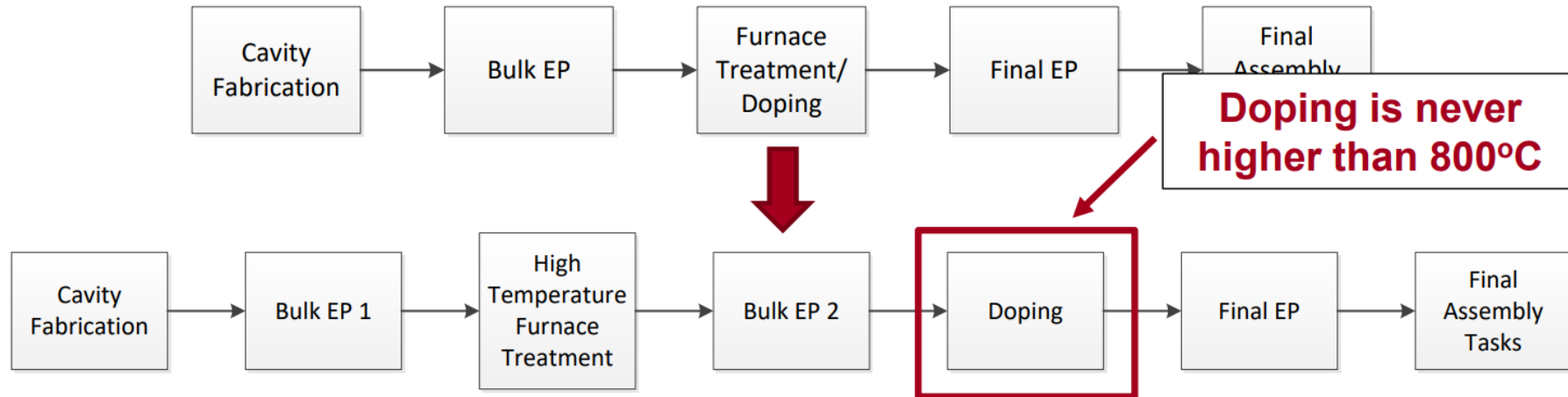


- 5 9-cell cavities prepared with 2/0 @ FNAL
- 4 9-cell cavity prepared with 3/60 @ JLab
- All cavities received the new “cold EP”
- 4 out of 5 of the **2/0 cavities** exceeded the HE specification by a wide margin
- 2 out of 4 **3/60 cavities** showed quench fields above 30 MV/m and $Q_0 > 4 \times 10^{10}$ at 21 MV/m

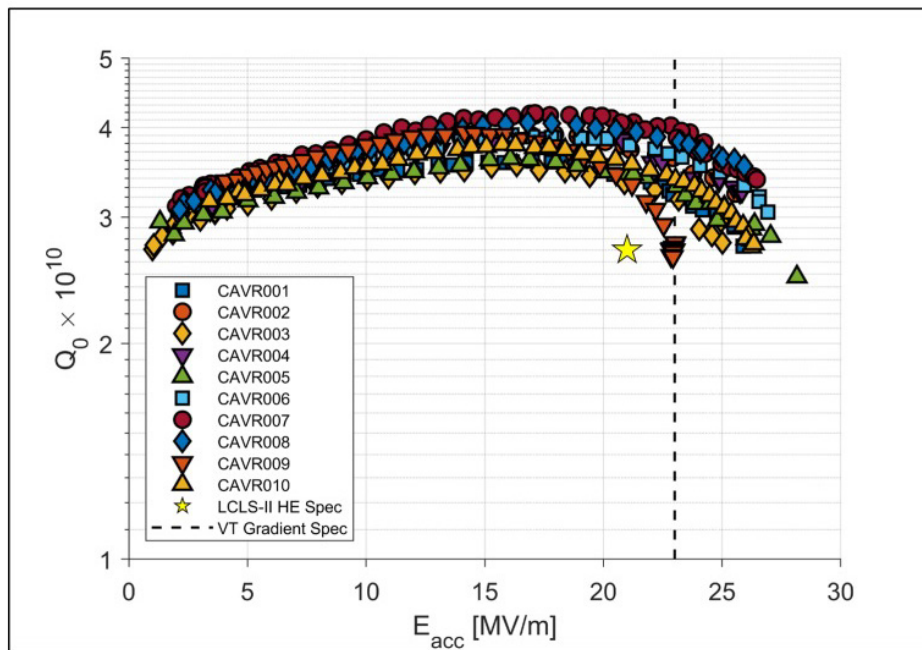
Changes from LCLS-II to LCLS-II-HE

Lessons learned from LCLS-II & HE R&D has resulted in two main changes to the cavity production process:

- Last portion of bulk EP and all of final EP must be done at “cold” temperatures
- An additional bulk EP and furnace step with increased RGA monitoring was added to reduce the chance of furnace contamination



Industry processed cavities for HE vCM



$\langle E_{acc} \rangle = 25.9$ MV/m
 $\langle Q_0 \rangle = 3.6 \times 10^{10}$

- 10 new 9-cells produced and processed by vendors with 2/0 recipe
- All cavities passed HE specification!
- Improvement of >3 MV/m compared to LCLS-II average

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First HE CM with 2/0 cavities: Fermilab vCM

- 8 of the 10 industrially produced cavities assembled in vCM at Fermilab
- Verification cryomodule → opportunity to:
 - Vet production capabilities of cavity vendors
 - Verify assembly and testing procedure at Fermilab
 - Evaluate operations of 1.3GHz CM at higher gradient

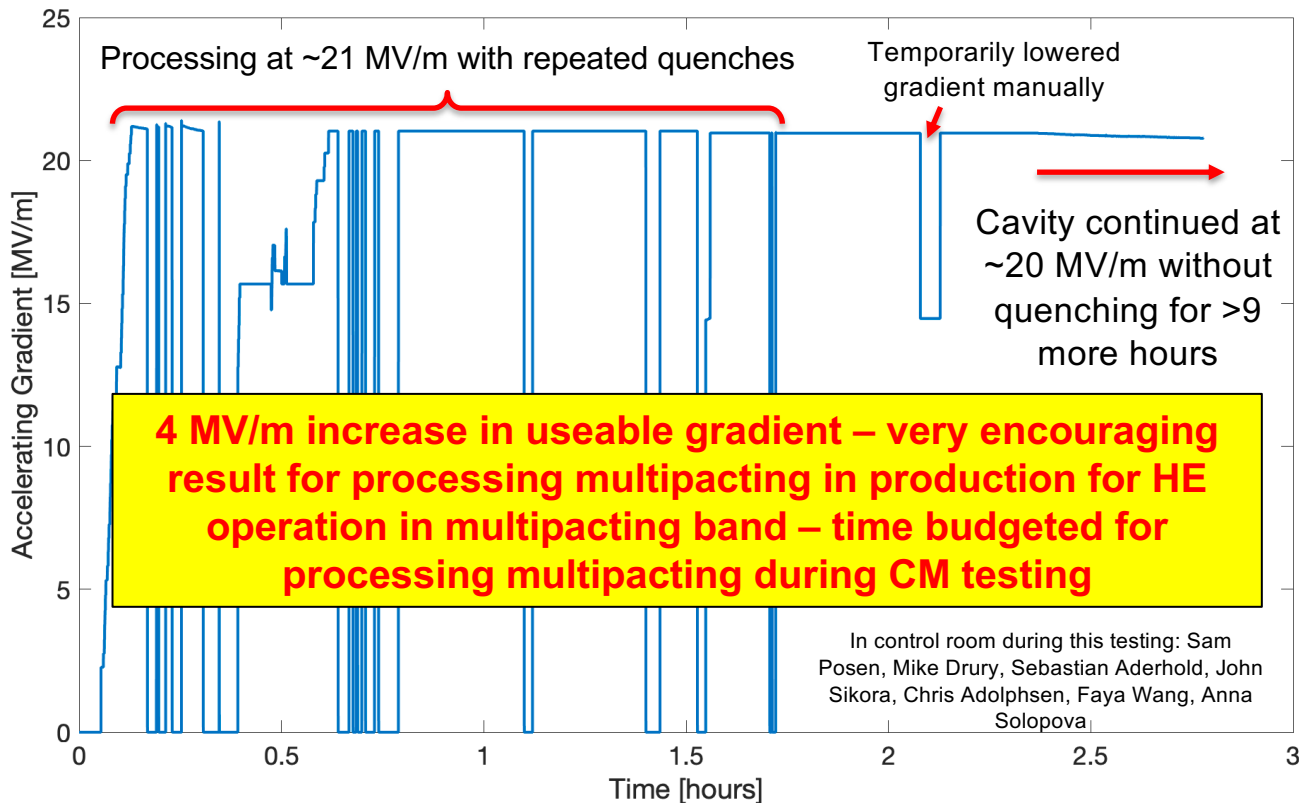


Evidence for MP-induced quenches

- In LCLS-II CMs, cases of usable gradients in the 17.5-18.5 MV/m range with max gradient closer to 20-21 MV/m
- Usable gradient requires 1 hour without quench, but regularly see cavities stable for many minutes then suddenly quench
- Critical for HE – to operate at 21 MV/m on average
- Supporting evidence for multipacting being source of “sporadic” quenches:
 - Sporadic quenches consistently observed only in multipacting band for TESLA shape ~17-24 MV/m
 - Quench coincides with burst of x-rays, suggesting electron activity
 - Processing (repeated quenching) helps to increase the gradient
 - No correlation is observed with endgroup temperature

Evidence for MP-induced quenches

- In L1
- max
- Usa
- stab
- Criti
- Sup
- S₁
- ~
- Q
- P₁
- N₁



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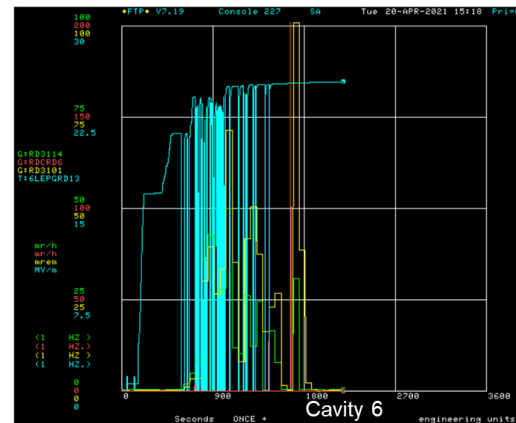
avities

enches:

ESLA shape

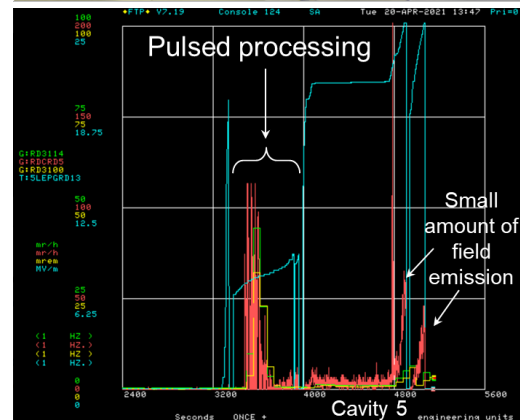
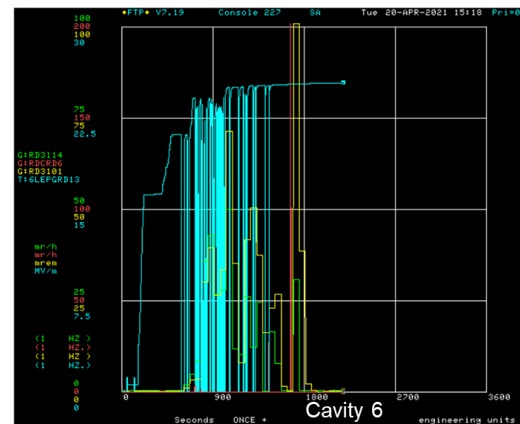
vCM performance

- Most cavities ramped to 16 MV/m without issue
- Rise to maximum gradient more eventful - lots of quenching, but quenches would process with time
- Every cavity had to do some quench processing
- Lots of radiation spikes, consistent with hypothesis that this is multipacting quenches
- Eventually cavities reach high gradient and are stable

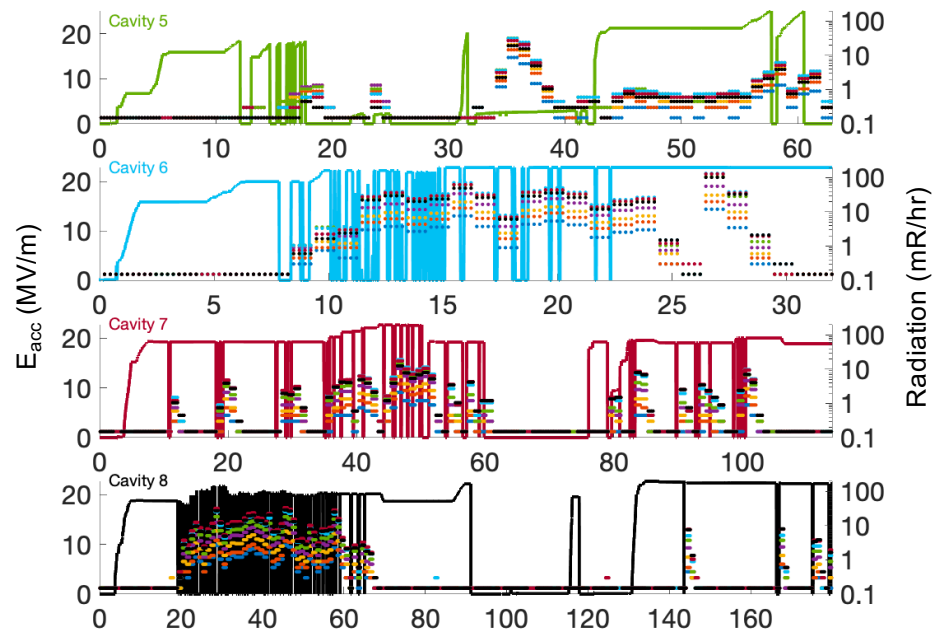
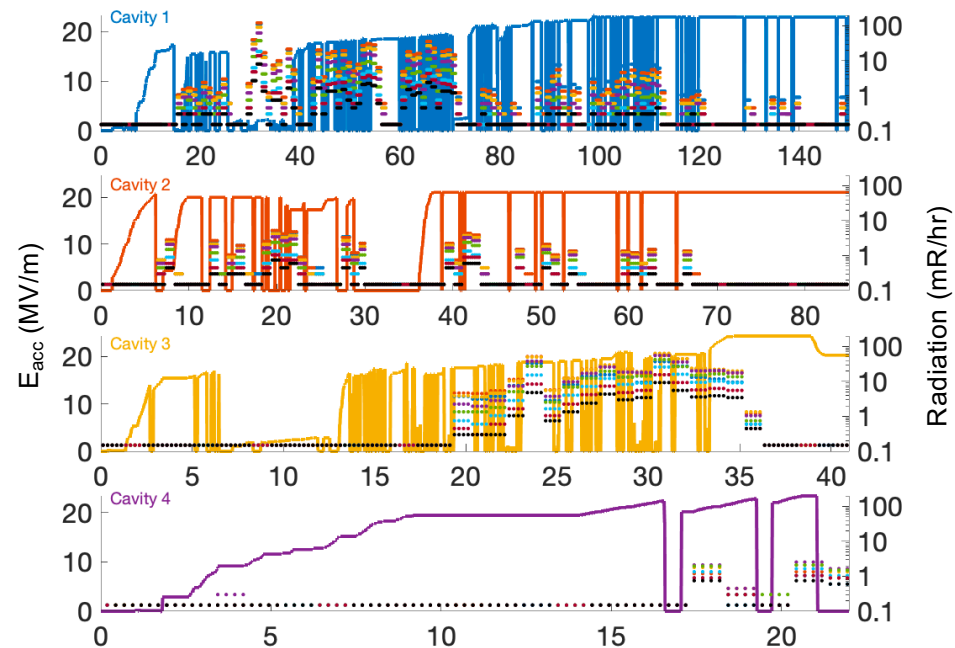


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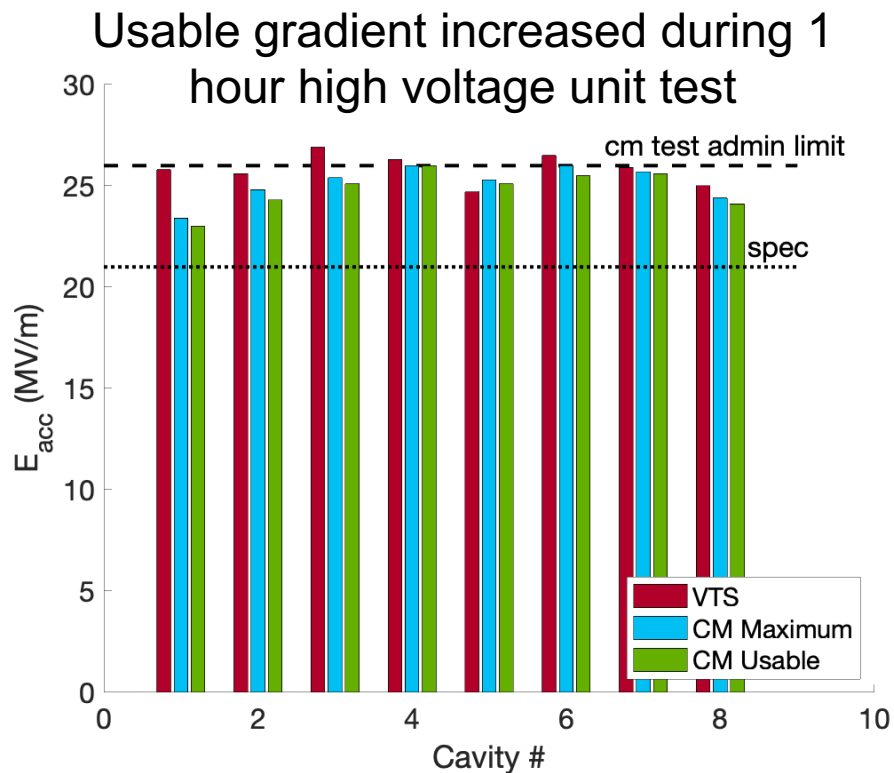
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- Cavity 5: only cavity that showed small amount of field emission, quickly processed. By the end of the vCM tests: no detectable FE up to the quench field or administrative limit!



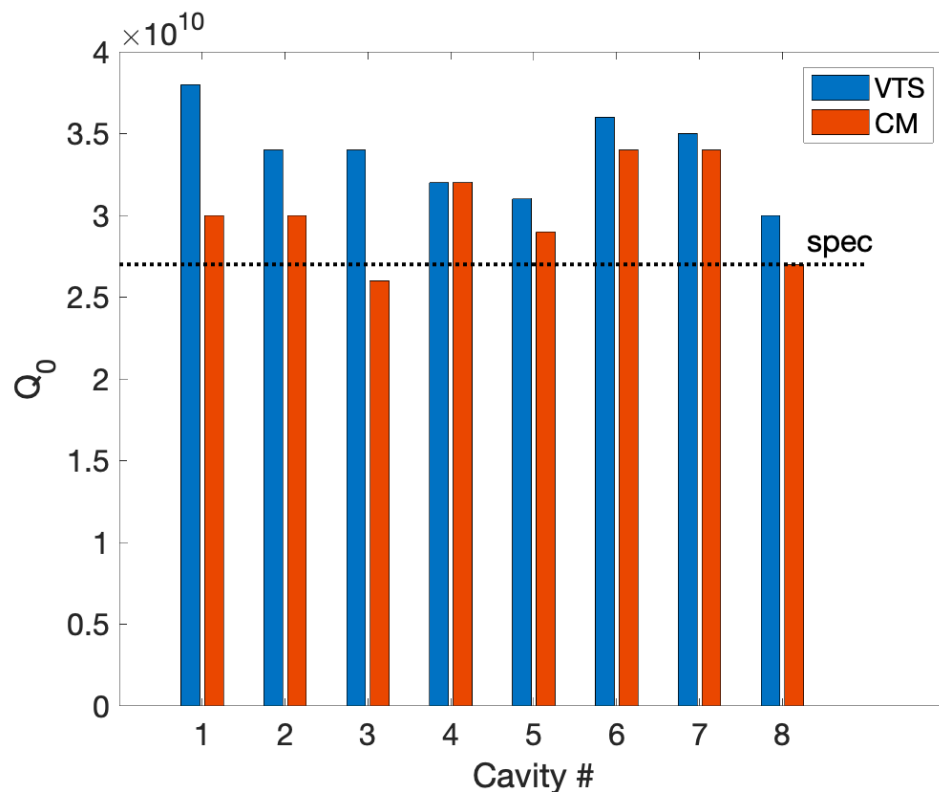
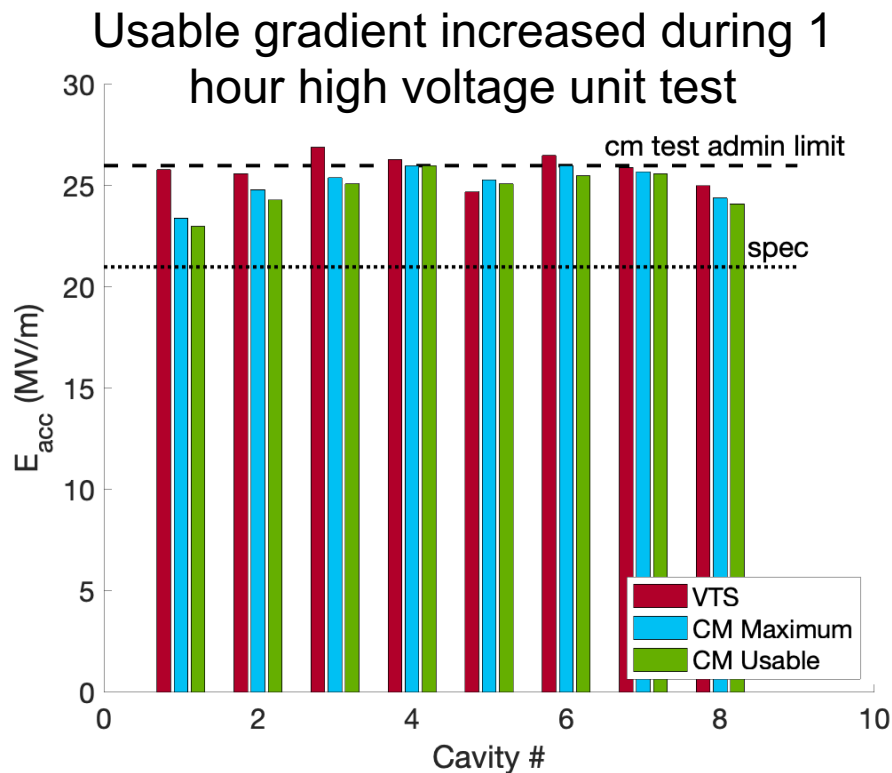
Power rise: summary of the eight cavities



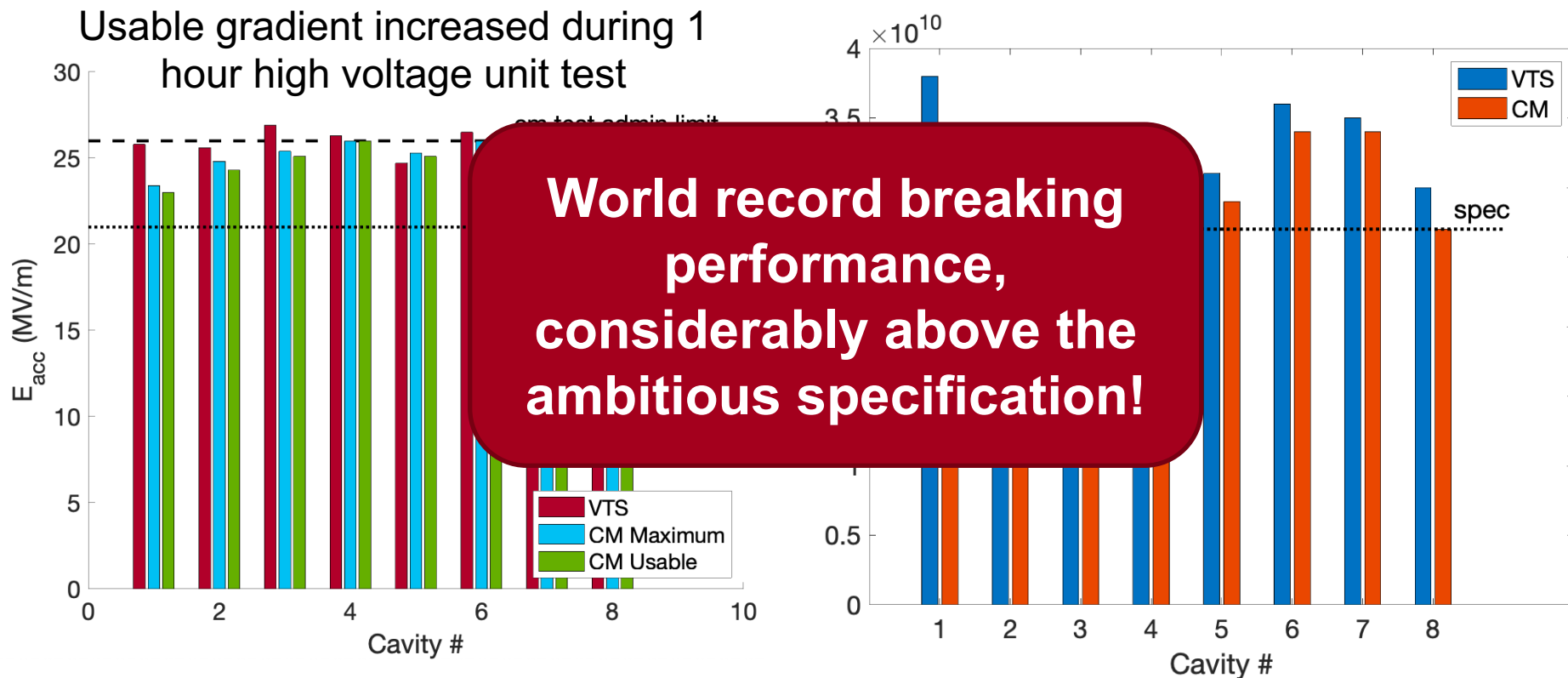
Cavity Gradient and Q_0 Performance summary



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Cavity Gradient and Q_0 Performance summary



Extended Unit Test

- Part of vCM test plan was an extended unit test – try to operate all 8 cavities in SELAP at nominal module voltage 173 MV
- SLAC operators travelled to Fermilab and took shifts so that at least one operator would be in the control room 24/7
- Duration: 12 days



SLAC visiting operators:

Sebastian Aderhold, Bob Legg, Janice Nelson, James Maniscalco, Lisa Zacarias

FNAL RF operators: Andrew Cravatta, Sam Posen

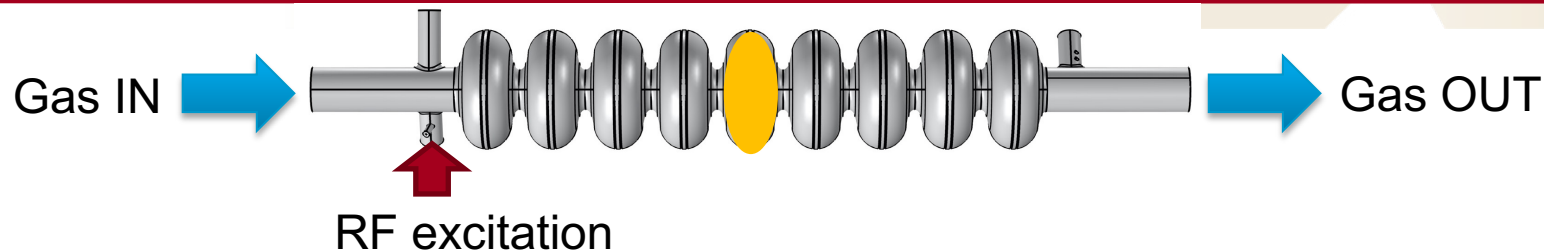
Conclusions on vCM performance

- Multiple doping recipes were investigated
 - High number and yield of 2/0 statistics motivate use of **2/0 for the cavity production**
- State-of-the-art industrial production for 9-cell SRF cavities was dramatically improved
- **vCM showed world record performance:**
 - demonstrated **total voltage of 200MV** (vs required 173MV)
 - Average **gradient >25MV/m** (vs required 20.8MV/m)
 - Average **$Q_0 > 3 \times 10^{10}$** (vs required 2.7×10^{10})

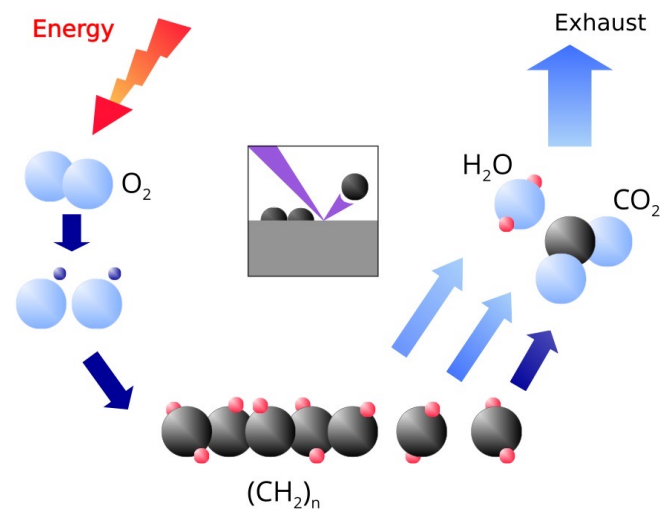
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Plasma processing for field emission mitigation (1)

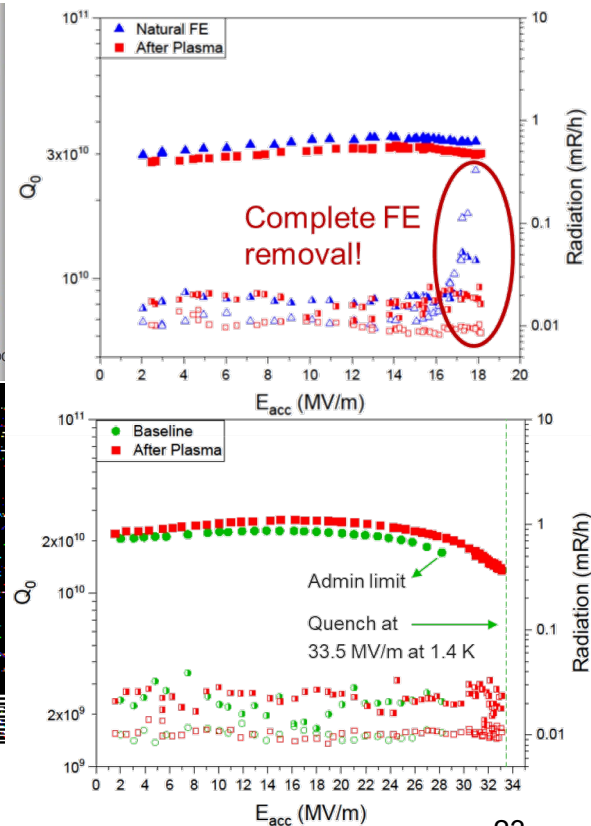
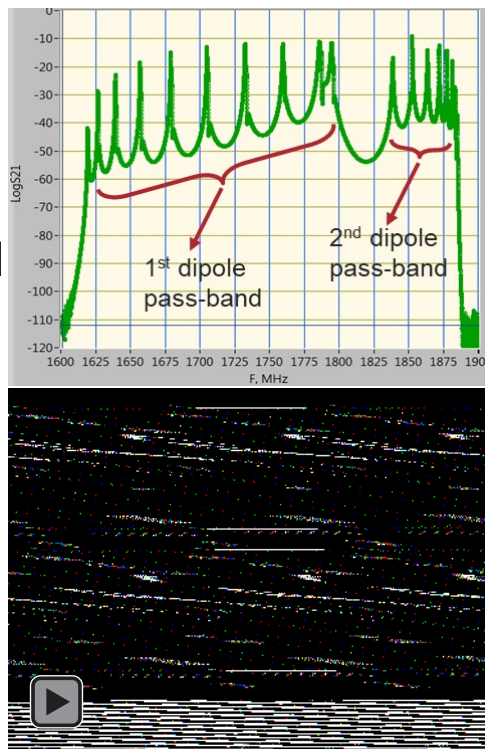


- Gas flow of Ne-O mixture (few % of O₂, mostly Ne) at p ~ 75-150mTorr;
- Once plasma is ignited, oxygen reacts with hydrocarbons;
- Reaction products (mostly CO, CO₂, H₂O) are pumped out;
- Work function increases, reducing FE.



Plasma processing for LCLS-II and LCLS-II-HE

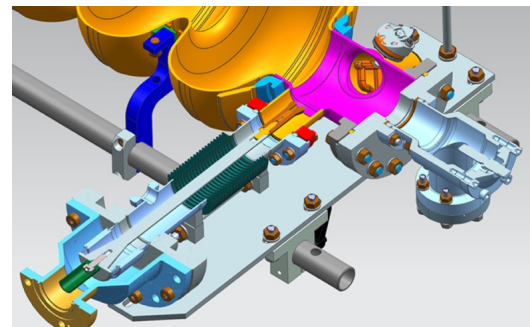
- Plasma ignition sequentially cell-by-cell using HOM modes and antennas
- Plasma is ignited in the central cell and moved to adjacent cells using a superposition of HOMs
- We demonstrated removal of hydrocarbon induced FE
- And no negative effect of plasma processing on N-doping: high Q_0 and quench field are preserved



Risk & mitigation analysis for vCM

- **Unstable pressure in the cryomodule:** new vacuum cart was procured & tested.
- **FPC ignition:** 'dummy' variable FPC installed on 9-cell cavity and subjected to plasma processing. No ignition in the FPC during processing.
- **Heating of HOM cables:** 100W for 30min do not degrade cables/connectors and do not cause excessive heating.
- **Heating of cavities:** vCM insulating vacuum was released.

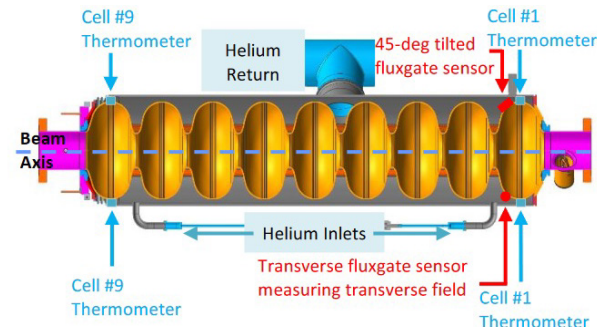
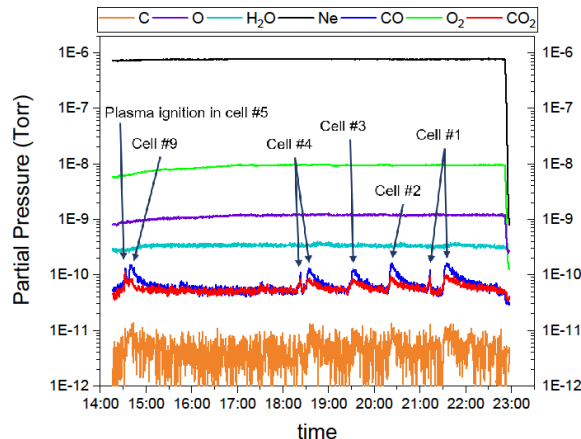
Cables and cavity temperature were monitored during plasma processing on vCM.



Plasma processing plan for vCM

Parameters monitored during plasma processing:

- Partial pressure of Ne, O₂, C, CO, CO₂, H₂O
- Pressure at the two ends of the cryomodule
- RF signals (forward & reflected power from HOM1, transmitted power from HOM2)
- Temperature of HOM1 cable connector, can and clamp
- Cavity temperature



Experimental systems: gas injection, vacuum & RF



Gas cart –
connected to
Feedcap side



Vacuum cart
– connected
to Endcap
side

RF system,
computers

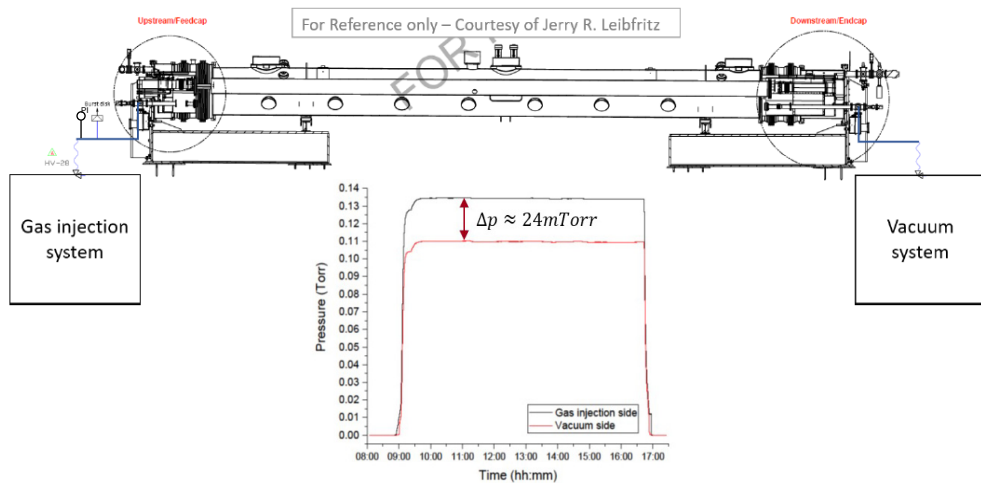


Connections between gas/vacuum systems and CM

Connections between vacuum/gas systems and vCM: conducted in cleanroom to minimize risk of particle contamination

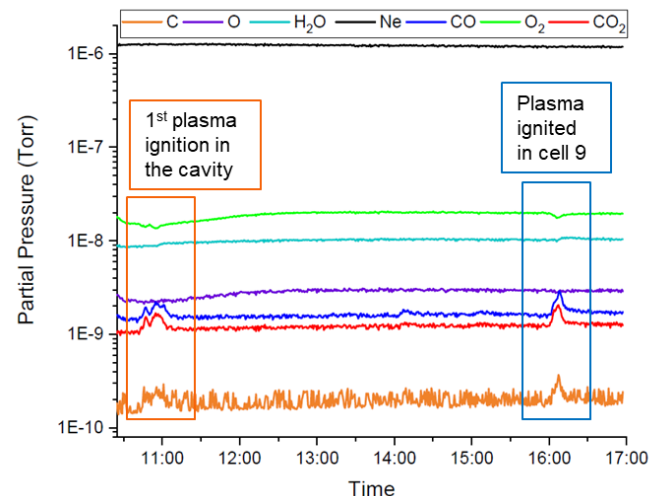


Plasma Processing applied to vCM (1)



Each morning the gas flow was established through the vCM

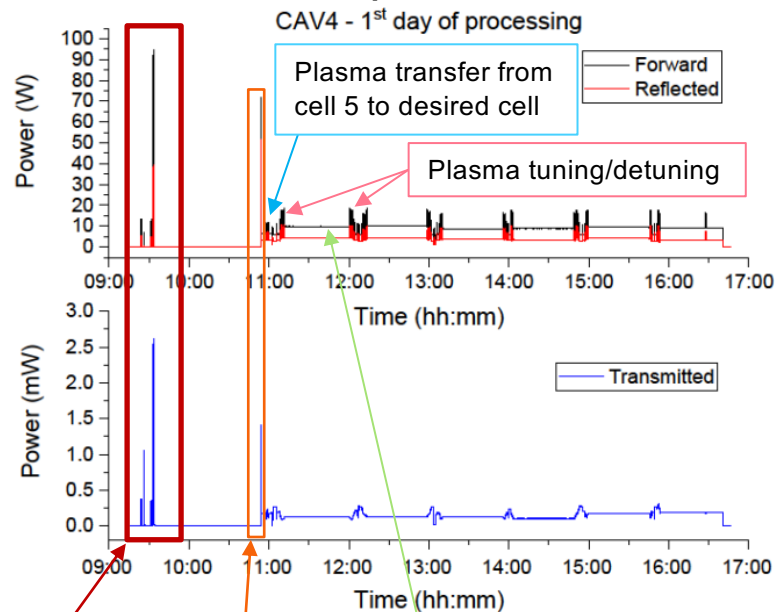
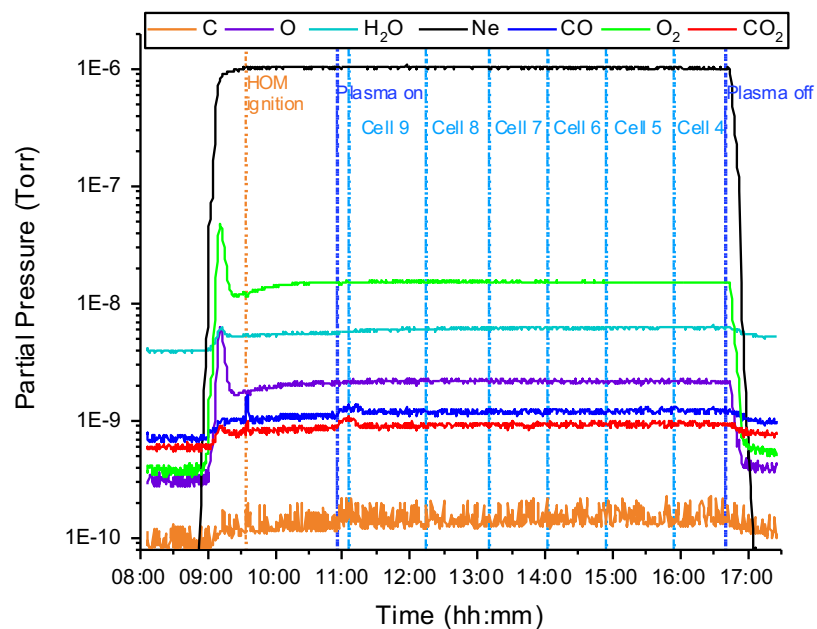
CAV1: 1st day of plasma processing



- Increase in CO, CO₂, C signals is observed along with decrease in O₂ signal
- Almost no by-products measured by RGA during 2nd day of plasma processing.

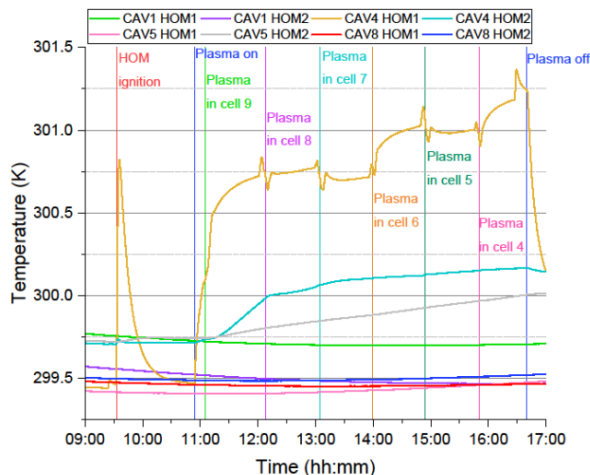
Plasma Processing applied to vCM (2)

Example of experimental data collected during plasma processing of CAV4. This includes a rare case of plasma ignition at the HOM coupler



Plasma Processing applied to vCM (3)

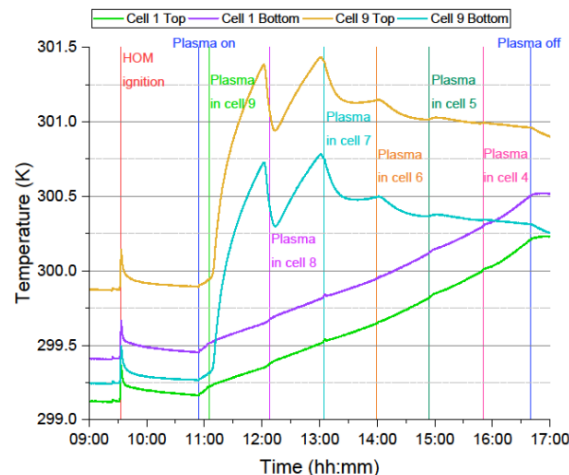
Example of experimental data collected during plasma processing of CAV4. This includes a rare case of plasma ignition at the HOM coupler



Temperature increase on:

- HOM1 cable < 2K
- HOM2 cable < 0.5K

During coupler ignition: 1.4K increase on HOM1 cable



Temperature increase on:

- Cell # 1 < 1.2K
- Cell # 9 < 1.6K
- During coupler ignition: 0.3K

Questions to address after plasma processing

RF test after processing to monitor changes in performance:

- Maximum gradient and usable gradient
- X-ray & Dark current
- Q-factor at 20.8 MV/m
- Check that cavities can sustain stable operation at 20.8 MV/m
- Time necessary to process multipacting

- Did plasma processing deteriorate cavity performance in any way?
- Did plasma processing have an impact on multipacting?

vCM performance before and after plasma processing

Cavity	Before Plasma Processing				After Plasma Processing			
	Max E_{acc} (MV/m)	Usable E_{acc} (MV/m)	Q_0 at 21 MV/m $\times 10^{10}$	MP quenches	Max E_{acc} (MV/m)	Usable E_{acc} (MV/m)	Q_0 at 21 MV/m $\times 10^{10}$	MP quenches
1	23.4	22.9	3.0	Yes	23.8	23.3	3.4	No
2	24.8	24.3	3.0	Yes	25.2	24.7	3.2	Yes
3	25.4	24.9	2.6	Yes	26.0	26.0	3.4	Yes
4	26.0	26.0	3.2	Yes	26.0	26.0	3.2	No
5	25.3	24.8	2.9	Yes	25.5	25.0	2.8	No
6	26.0	25.5	3.4	Yes	26.0	26.0	3.2	Yes
7	25.7	25.2	3.4	Yes	25.9	25.4	3.3	Yes
8	24.4	23.9	2.7	Yes	24.7	24.2	2.6	No
Average	25.1	24.7	3.0		25.3	25.1	3.1	
Total	209	205			210	208		

RF test after plasma processing demonstrated that:

- **vCM performance** are **preserved**
- Plasma processing did not introduce any contamination: **vCM still FE-free**

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**Plasma processing
procedure is fully
validated!**

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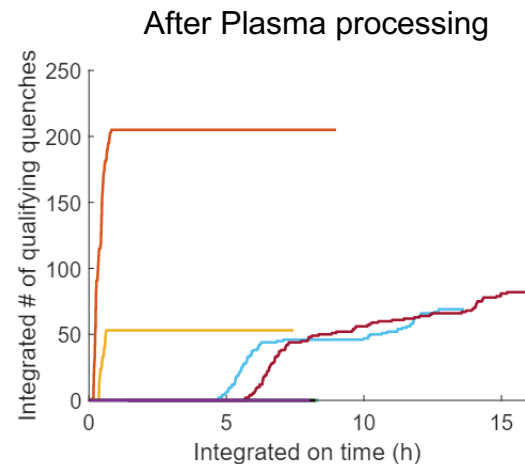
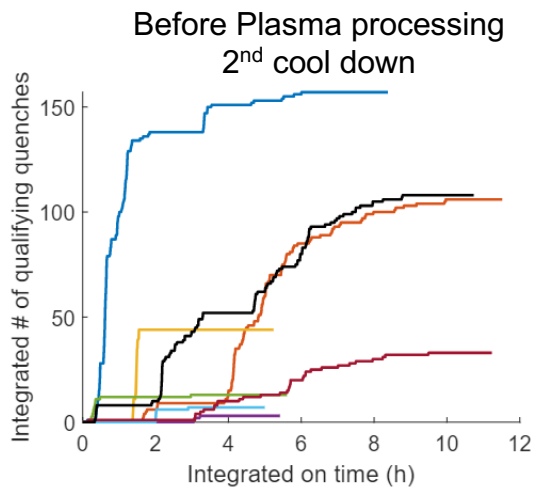
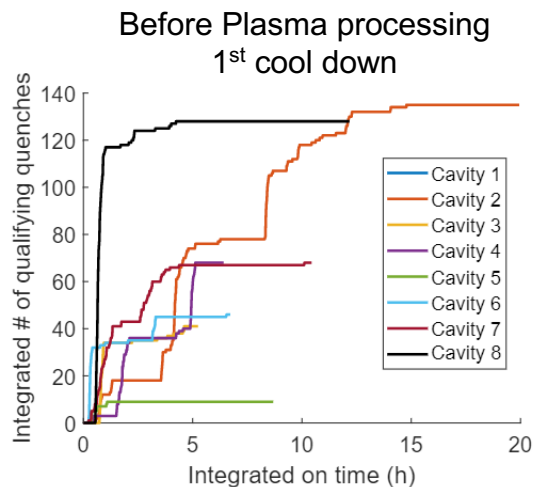
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vCM performance before and after plasma processing



Plasma processing **can eliminate multipacting:**

- the 4 plasma processed cavities do not exhibit any MP quench, contrary to the other 4 cavities

We could **address both FE and MP in situ at the same time**

Cavity	Multipacting Quenches		
	Before plasma Processing		After Plasma Processing
	1 st cooldown	2 nd cooldown	
1	/	157	0
2	135	106	205
3	41	44	53
4	68	3	0
5	10	16	0
6	46	7	69
7	68	33	82
8	128	108	0

Conclusions

- Plasma processing was already validated in test bench setup on N-doped LCLS-II 9-cell cavities
- Through this test the **procedure for LCLS-II and HE was fully validated on a CM**: plasma processing has the potential of not only reducing field emission but also fully eliminating multipacting in cryomodules.
- **Multipacting mitigation through plasma processing** could:
 - significantly reduce cryomodule testing time
 - decrease the accelerator commissioning time and cost (for example: cavities can require up to 1 day of processing in case of severe multipacting, with an average of 2h for cavity)
 - increase reliability during operations



Thank you!

Timeline & Procedure

- No cavity with FE in vCM → processed 4 instrumented cavities (CAV1, CAV4, CAV5, CAV8)
- Processing order: started from CAV1 (closest to vCM end, gas injection side) and proceeded to CAV4, CAV5, CAV8 following the direction of gas flow.
- Processing time: 1.5-2 hours/cell -> 13.5-18 hours/cavity
- Plasma ignition procedure:
 - Plasma is ignited in cell # 5 (requires $\approx 60-70\text{W}$)
 - Plasma is transferred to cell # 9 (going through cell # 6, 7, 8)
 - Plasma density is maximized in cell # 9 and cell processed for 45 min
 - Plasma density is detuned and plasma is transferred to cell #8
 - Procedure is repeated for each cell

Conclusions and lessons learned from Plasma processing on vCM

- Pressure in vCM: stable across the cavity string; pressure drop between 2 sides was consistently around 24mTorr
- Higher order modes are not tuned: great variation in frequency and dipole splitting. Nevertheless, plasma ignition with HOMs demonstrated to be a robust method that allowed to process all 4 cavities
- Plasma processing is confirmed as a **promising technique to reduce field emission and multipacting levels in-situ in LCLS-II HE CMs** increasing the usable gradient

HOM ignition

